

Search Ad Optimization — Research Update

February 26, 2026

Agenda

- ① **SVD dimension sweep** — optimal number of embedding dimensions
- ② **Minimum budget constraint** — eliminating low-spend keywords
- ③ **SCIP vs. Gurobi** — solver performance comparison

SVD Dimension Sweep

Experiment design

Goal: Find the optimal SVD dimensionality for MiniLM embeddings (384-d $\rightarrow$ k -d).

Setup:

- Courses: Sys Think and ML; backtest Dec 1–31, 2025.
- $k \in \{10, 20, 50, 100, full\}$; $full = 384$ (no SVD).
- SVD fitted daily on historical data only (no leakage).

Old Eval vs. Oracle Eval

Evaluation model used to score backtest clicks

Oracle model: CV over $k \in \{10, 20, 50, 100, 384\}$; selects $k=384$ for both courses.

- This model evaluates the number of clicks each cost allocation produces in the backtest.
- Old eval: single XGBoost with fixed $k=50$ SVD, trained on all data.
- Oracle CV MSE: 1.73 (Sys Think), 2.62 (ML) — lower than old eval (1.81, 2.81).
- Old exp results were overstated due to SVD data leakage and worse evaluation model.

SVD Sweep Results — Sys Think

Dec 1–31 backtest, \$357 budget

Experiment	Optimized (daily avg)					Improvement		
	Purch	Clicks	Cost	Clicks/\$	Kws	Purch	Clicks	Clicks/\$
Old exp (old eval)	0.34 ± 0.01	265.2 ± 10.5	357.05	0.743	119	+31.7%	+116.9%	+101.5%
Old exp (Oracle)	0.30 ± 0.01	189.9 ± 8.0	357.05	0.532	119	+18.1%	+56.1%	+45.1%
$k=10$	0.23 ± 0.01	138.2 ± 5.5	357.01	0.387	92	-6.5%	+15.4%	+5.2%
$k=20$	0.28 ± 0.01	171.2 ± 6.8	357.08	0.479	98	+11.7%	+42.9%	+30.3%
$k=50$	0.29 ± 0.01	199.2 ± 7.2	356.93	0.558	96	+17.6%	+66.3%	+51.7%
$k=100$	0.28 ± 0.01	178.7 ± 8.0	356.87	0.501	94	+13.7%	+49.2%	+36.1%
Full (384)	0.34 ± 0.01	196.7 ± 12.4	357.09	0.551	122	+34.3%	+61.7%	+50.3%

Baseline: Purch: 0.25 ± 0.01 , Clicks: 119.8 ± 2.8 , Cost: $\$325.62 \pm 10.12$, Clicks/\$: 0.368, Kws: 36 ± 1 .

- Old exp's results worsen with new evaluation model - similar to $k=50$ results.
- Full embeddings maximize purchases (**+34.3%**).

SVD Sweep Results — ML

Dec 1–31 backtest, \$353 budget

Experiment	Optimized (daily avg)					Improvement		
	Purch	Clicks	Cost	Clicks/\$	Kws	Purch	Clicks	Clicks/\$
Old exp (old eval)	0.15 ± 0.01	122.6 ± 10.0	353.94	0.346	410	+30.2%	-16.3%	-21.4%
Old exp (Oracle)	0.11 ± 0.01	85.4 ± 4.4	353.94	0.241	410	-6.4%	-38.4%	-42.2%
$k=10$	0.12 ± 0.01	104.0 ± 5.0	353.95	0.294	358	+8.2%	-24.9%	-29.6%
$k=20$	0.12 ± 0.01	98.1 ± 3.9	353.92	0.277	340	+5.8%	-29.2%	-33.5%
$k=50$	0.11 ± 0.01	93.1 ± 3.5	353.97	0.263	353	+0.1%	-32.8%	-37.0%
$k=100$	0.12 ± 0.01	94.6 ± 4.2	353.92	0.267	382	+5.0%	-31.5%	-36.0%
Full (384)	0.14 ± 0.01	110.4 ± 8.4	353.95	0.312	472	+22.8%	-20.6%	-25.3%

Baseline: Purch: 0.11 ± 0.00 , Clicks: 138.5 ± 4.7 , Cost: $\$332.00 \pm 14.27$, Clicks/\$: 0.417 , Kws: 38 ± 1 .

- Full embeddings maximize purchases (+22.8%).

Regional Budget Allocation

Spend share by region

Sys Think

	USA	A	B
Actual	65.7%	24.3%	10.1%
Old exp	35.9%	35.2%	28.9%
SVD Full	41.4%	40.6%	18.0%

ML

	USA	A	B
Actual	75.3%	15.1%	9.6%
Old exp	93.2%	6.8%	0.0%
SVD Full	95.9%	4.1%	0.0%

- Sys Think: optimizer shifts spend from USA toward cheaper regions A and B.
- ML: optimizer concentrates nearly all spend in USA — explains click loss.

Policy Model Over-Prediction

Gurobi-predicted clicks vs. Oracle-evaluated clicks

Course	Policy Clicks	Oracle Clicks	Over-prediction
ML	8,839	3,202	2.76×
Sys Think	9,850	6,099	1.62×

- Policy clicks = $\sum_i (\hat{f}_i^{\text{policy}}(\mathbf{x}_i^*) - \hat{f}_i^{\text{policy}}(\mathbf{0}))$.
- Oracle clicks = $\sum_i (\hat{f}_i^{\text{oracle}}(\mathbf{x}_i^*) - \hat{f}_i^{\text{oracle}}(\mathbf{0}))$.
- **How can we improve?** Identify where over-prediction is high and constrain the optimizer.

Diagnosing Over-Prediction — By Daily Spend

Low daily spend drives over-prediction

ML		
Daily Spend	<i>N</i>	Ovr%
<\$0.50	4,735	925%
\$0.50–1	5,934	147%
\$1–5	2,980	109%
\$5–10	36	73%
>\$10	1	36%

Sys Think		
Daily Spend	<i>N</i>	Ovr%
<\$0.50	108	≥100%
\$0.50–1	2,601	1,208%
\$1–5	599	268%
\$5–10	24	34%
>\$10	451	48%

N = number of (keyword, match type, region) rows in optimal solution across 31 backtest days.

Ovr% = $(\sum \text{Policy clicks} / \sum \text{Oracle clicks} - 1) \times 100$.

- Daily spend <\$1 is rare in the historical data (< 25%).
- Keywords with <\$1 daily spend account for the majority of over-prediction.

Min-Spend \$1 Constraint — Results

Dec 1–31 backtest, Full embeddings ($k=384$)

Sys Think:

Experiment	Optimized (daily avg)					Improvement		
	Purch	Clicks	Cost	Clicks/\$	Kws	Purch	Clicks	Clicks/\$
No floor	0.34 ± 0.01	196.7 ± 12.4	357.09	0.551	122	+34.3%	+61.7%	+50.3%
Min \$1	0.36 ± 0.01	215.4 ± 12.0	356.81	0.604	81	+42.5%	+77.1%	+64.5%

ML:

Experiment	Optimized (daily avg)					Improvement		
	Purch	Clicks	Cost	Clicks/\$	Kws	Purch	Clicks	Clicks/\$
No floor	0.14 ± 0.01	110.4 ± 8.4	353.95	0.312	472	+22.8%	-20.6%	-25.3%
Min \$1	0.19 ± 0.01	151.8 ± 6.2	353.89	0.429	266	+67.3%	+9.5%	+2.8%

- Improvement in both courses, for both purchases and clicks.
- Active keywords reduced: $122 \rightarrow 81$ (Sys Think), $472 \rightarrow 266$ (ML).

SCIP vs. Gurobi

Solver performance comparison

Model size: >1M variables, >3M constraints.

Solver	Cores	Mem (GB)	Solve Time
SCIP	8	64	>10 hours (got to 9% optimality gap)
Gurobi	16	128	~10 minutes

- Gurobi is orders of magnitude faster for our MILP formulation (though not a fair comparison - will redo).
- SCIP is open-source but may be impractical at this problem scale.

- **Full embeddings (no SVD)** perform best for both courses.
- Policy model over-predicts clicks by 1.6–2.8 $\times$, driven by low-spend keywords.
- **Min-spend \$1 floor** eliminates the worst over-prediction:
 - Sys Think: purchases +**42.5%**, clicks +**77.1%**.
 - ML: purchases +**67.3%**, clicks +**9.5%**.
- **Gurobi** solves the model ($>1\text{M}$ variables) in ~ 10 min; SCIP did not finish in >10 hours.